Unit 4 Files and Directories

Course code LX13
Linux Basics

Objectives

After completing this unit, students should be able to:

- Describe the different file types
- Describe the Linux file system structure
- Use both full and relative path names in a file specification
- Create, delete and list directories
- Use the `touch` command to create an empty file

A File

A file is:

- A **collection of data**
- A **stream of characters** or a "**byte stream**"
- **No structure** is imposed on a file by the operating system

File Types

- Ordinary
  Text or **code data**
  No particular internal format

- Directory
  A **table of contents**
  A list of the files within that directory

- Special Files
  Represent **hardware** or **logical devices**
  Found in directory called **/dev**
  For example: **/dev/lp0** (printer), **/dev/null** (null device)

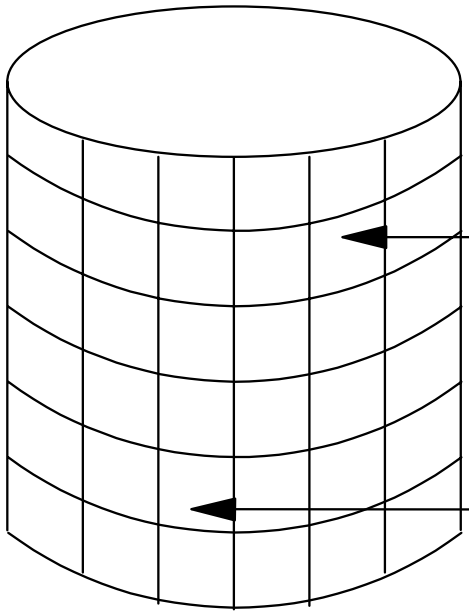
Directory Contents

Directory

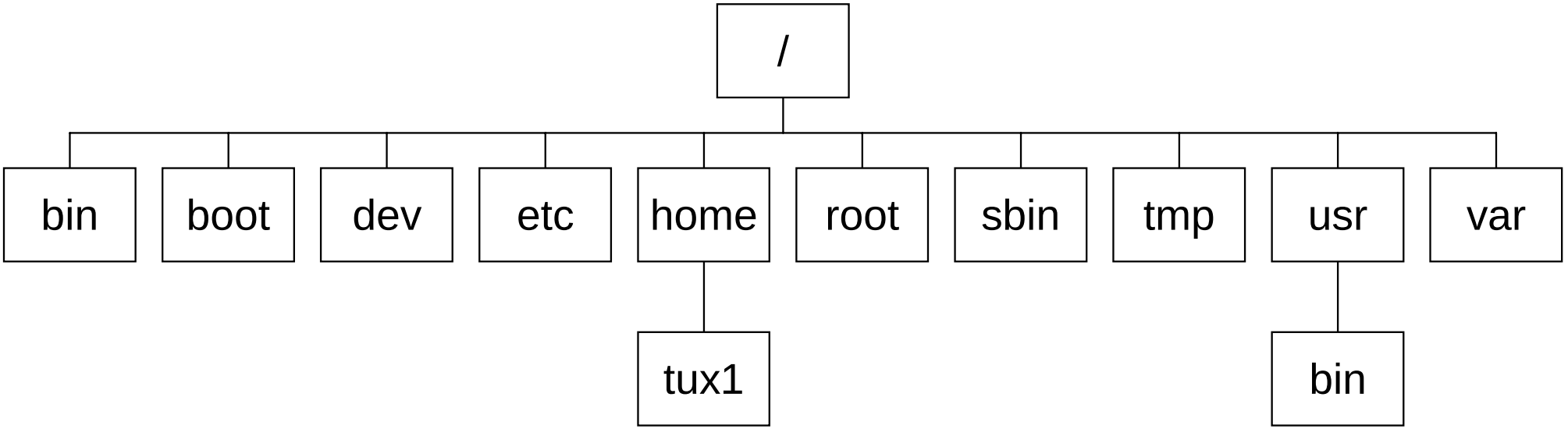
Name	inode
subdir	4
myfile	10

Inode Table

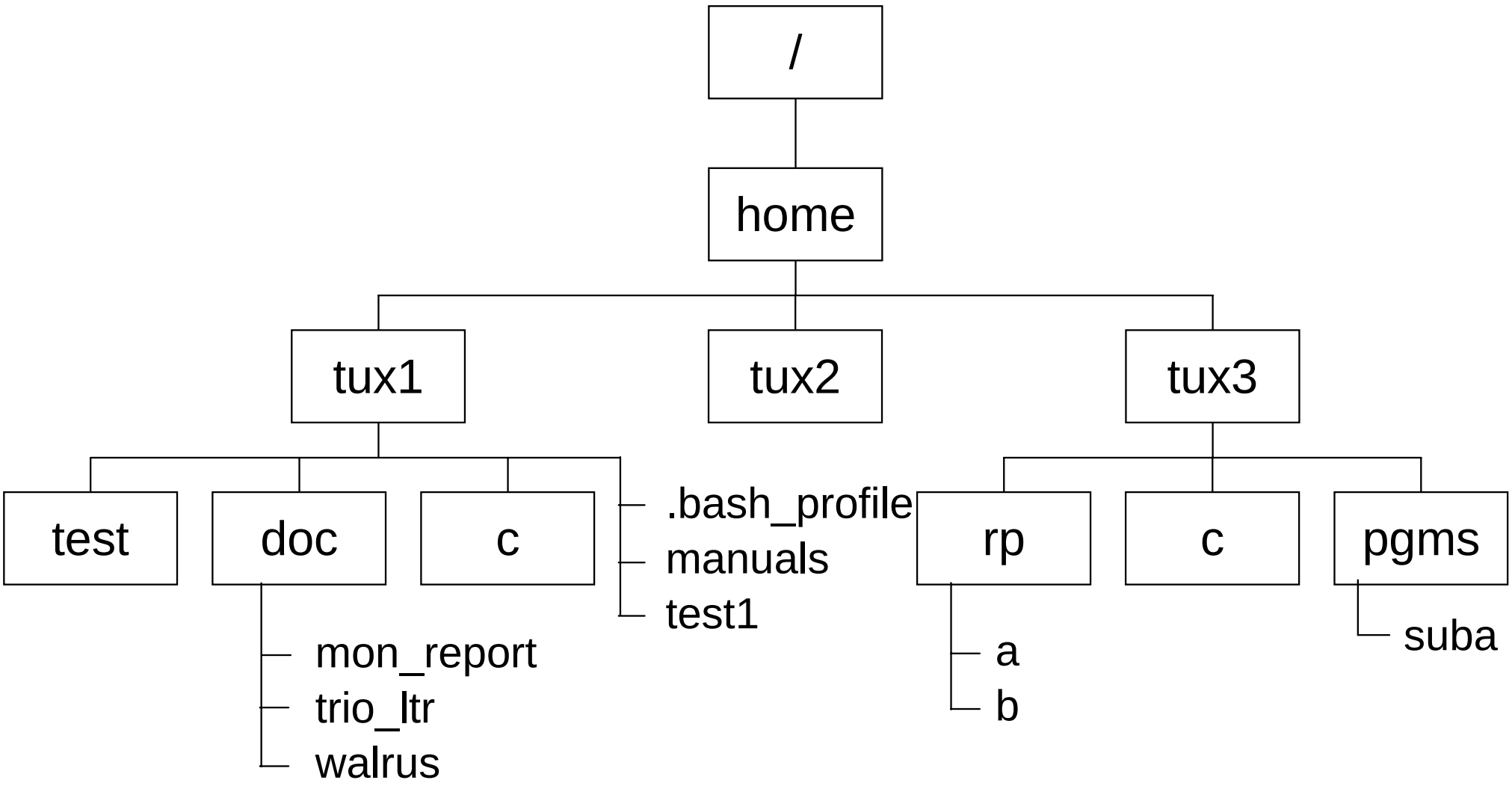
#	Type	Mode	Link	User	Group	size	date	loc
4	d	755	2	team01	staff	1024	Feb 17 12:55	
10	-	644	1	team01	staff	314	Feb 16 16:07	



Hierarchical Structure



Example Directory Structure



Pathnames

- A sequence of file names, seperated by slashes (/), that describes the path, or route, the system must follow to locate a file in the system

- Two types:
  - Full - Start from / (the **root** directory)
  - Relative - Start from the current directory

- Examples: (working directory is /home/team01)

/home/tux1/doc/mon_report	(full)
doc/mon_report	(relative)
../tux3/pgms/suba	(relative)
./test1	(relative)

Where Am I ?

- The **p**rint **w**orking **d**irectory can be used to find out what your current working directory is:

```
$ pwd
/home/tux1
```

Change Current Directory

- With the **c**hange **d**irectory command:

  $ **cd dir_name**

- If you want to move from /home/tux1 to /home/tux1/doc:

  $ **cd doc** (relative)

  or

  $ **cd /home/tux1/doc** (full)

  $ **cd** Go to **your** home directory

  $ **cd ..** Go to the parent directory of the current directory (one step up)

Linux Filenames

- Should be descriptive of the content
- Should use only alphanumeric characters
  UPPERCASE, lowercase, number, @, _
- Should not include embedded blanks
- Should not contain shell metacharacters
  *** ? > < / ; & ! [ ] | \ ' " ( ) { }**
- Should not begin with + or - sign
- Should not be the same as a system command
- Are case sensitive
  rainbow and **Rainbow** represent different files
- Filenames starting with a **.** are hidden from the normal ls command. (ls -a also shows the hidden files)
- The maximum number of characters for a filename is 255

Create Directories

- With the make directory command:
  `$` **mkdir dir_name**

- To create the directory **doc**, as a subdirectory of **/home/tux1**:

```
$ mkdir /home/tux1/doc
  or
$ cd /home/tux1
$ mkdir doc
```

Removing Directories

- With the remove directory command:

```
$ rmdir dir_name
```

- For this command:
  1. dir_name must be empty
  2. dir_name must not be the current directory

```
$ pwd
/home/tux1
$ rmdir doc
rmdir: doc: Directory not empty
$ rmdir test
$
```

Working with Multiple Directories

- Create multiple directories simultaneously:
  ```
  $ mkdir -p dir1/dir2/dir3
  ```

- Remove multiple directories simultaneously:
  ```
  $ rmdir -p dir1/dir2/dir3
  ```

List the Contents of Directories

- With the **ls** command:
  `$ ls dir_name` (dir_name is optional)

- To list contents of your home directory

```
$ pwd
/home/team01
$ ls
c  doc  manuals  test1
```

- To list **all** files, including hidden (.) files:

```
$ ls -a
.     .bash_history .bash_profile doc       test1
..    .bash_logout  c             manuals
```

List the Contents of Directories *(2)*

- To list all files and directories to the end of the directory tree:

```
$ ls -R
c    doc manuals   test1

./c:

./doc:
mon_report trio_ltr  walrus
```

Coloring the Output

- The outout of ls can be colorized.
- Use the --color option

- Some colors used by ls
  - ordinary files default
  - directory blue
  - executable green
  - special files yellow

```
$ ls --color
```

Long Listing of Files

- The ls command with the **-l** option can be used to obtain more information about files and directories.

```
$ ls -l
drwxrwxr-x  2 team01 team01  1024 Feb 17 14:07  c
drwxrwxr-x  2 team01 team01  1024 Feb 17 14:07  doc
-rw-rw-r--  1 team01 team01   314 Feb 17 14:08  test
```

- The -i options displays the inode number of the file

```
$ls -li
5 -rw-rw-r-- 1 team01 team01 314 Feb 17 14:08 lx
```

Displaying Directory Information

- Normally ls shows the contents of a directory when you specify a directory name as the parameter of ls. You can prevent this with the **-d** option.

```
$ cd /home/tux3
$ ls -li rp
10 -rw-rw-r-- 1 team01 team01 302 Mar  5 15:29 a
11 -rw-rw-r-- 2 team01 team01 301 Mar  5 15:30 b

$ ls -l di rp
5 drwxrwxr-x 2 team01 team01 1024 Feb 17 14:07  rp
```

The touch Command

- The touch command updates the access and/or modification times of a file.
- The touch command can also be used to create zero-length files

```
$ ls -l
-rw-rw-r-- 1 team01 team01  512 Feb 24 11:10 docs

$ touch docs new
$ ls -l
-rw-rw-r-- 1 team01 team01 512 Mar  5 15:37 docs
-rw-rw-r-- 1 team01 team01   0 Mar  5 15:37 new
```

Checkpoint

1. Using the tree structure shown earlier and using **/home** as your current directory, how would you refer to the **suba** file in the **pgms** directory using both full and relative path names?

2. When specifying a path name, what is the difference between the . and the ..?

Checkpoint *(2)*

3. What will the `cd ../..` command do?

4. What conditions have to be satisfied in order for rmdir to complete succesfully?

Checkpoint *(3)*

5. Match the various options of the **`ls`** command with their functions.

a. -a __ Provides a long listing
b. -i __ Will list hidden files
c. -d __ Lists subdirectories and their contents
d. -l __ Displays the inode number
e. -R __ Displays information about a directory

6. Which of the following are valid file names

a. 1
b. AbCdEe
c. -myfile
d. my_file
e. my file
f. .m.y.f.i.l.e.
g. my*file
h. .my_file

Unit Summary

- There are three types of files:
  - Ordinary
  - Directory
  - Special

- The Linux file system structure is a hierarchical tree.

- Files are accessed using either full or relative path names. A full pathname always begins with a ***/***.

- The following commands can be used with directories:
  `pwd, cd, mkdir, rmdir, touch` and `ls`